Project Name: Vistelligence (An AI-Based Visa Predictor)

Team Members: Kubra Noor, MH Sherharyar, Hina Ijaz
Advisor: Ms. Hira Asim
Department: Faculty of IT & CS

Group ID:F24SE055

Abstract

Vistelligence is an AI-powered platform that simplifies visa applications by offering tailored visa, university, and country recommendations. It predicts visa approval chances, assists in interview preparation, and provides budgeting tools. The system also includes a peer-support community to reduce applicant stress. With real-time data and AI algorithms, it empowers users to make confident decisions about studying, working, or investing abroad.

Introduction

Visa processes are often overwhelming and full of uncertainty. Most existing platforms offer one-size-fits-all advice that leaves users confused and unprepared. **Vistelligence** changes that narrative through an AI-powered web application that analyzes each user's unique situation education, finances, career goals and provides **personalized visa guidance**.
At the core of our system is a **RAG-based AI model** (Retrieval-Augmented Generation), where users input prompts like their destination preference or academic background and the system **retrieves relevant knowledge** and **generates tailored visa, country, and university suggestions**.
The platform offers targeted interview preparation and hosts a **forum** for peer support and shared experiences. Vistelligence is designed for students, professionals, and investors who are navigating the global mobility maze

Methods and Materials

- Built with **React** and **Node.js**
- **RAG-based AI model** utilizes user profile data (education, funds, preferences, etc.)
- REST APIs used for seamless front-end and back-end communication
- Integrated **MongoDB** for flexible and scalable data management
- Includes **Admin Panel** and **User Profile Dashboard** modules

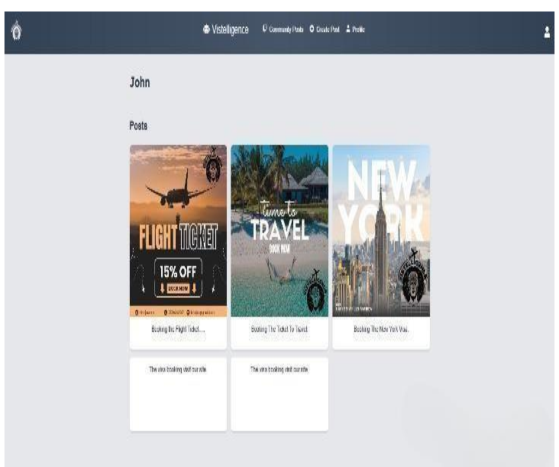


Figure 1. User Profile

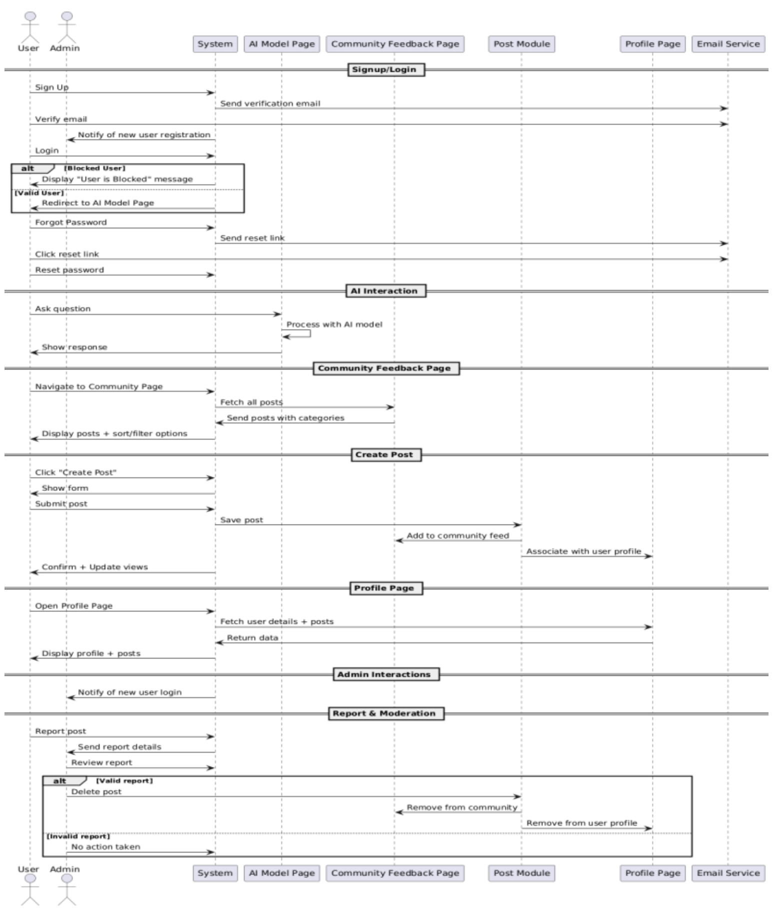


Figure 2. Sequence diagram

Discussion

Vistelligence is an AI-powered visa guidance platform. When a user logs in and submits a prompt like "Hey, I am 23-year-old , fresh graduate of Software engineering Which visa suits me for Canada?" our **RAG-based AI model** springs into action. It first retrieves relevant visa data from a knowledge base, then generates a **context-aware, tailored response** based on the user's profile (education, finances, goals).
The platform's workflow connects all major modules. The **front-end** handles login, profile setup, and community posts. The **back-end** manages data storage and interactions between modules. Our **AI model** processes user queries and delivers meaningful guidance. The **admin panel** allows moderators to manage users and review posts to keep the platform safe and useful.
Behind the scenes, every action from posting to AI recommendations is handled through a well-coordinated sequence of system responses. This makes Vistelligence not just smart, but **user-centered, scalable, and reliable** for real-world use.

Results

- **20 test cases executed**; performance, load, and remaining testing will be conducted in Phase 4
- **Majority passed**, including registration, login, and AI prediction functionalities
- **Minor UI and email handling issues** are still pending
- **AI Recommendation Engine** and **Cost Estimator** are **partially completed**
- **Admin Panel** and **Community Forum** are **fully functional**

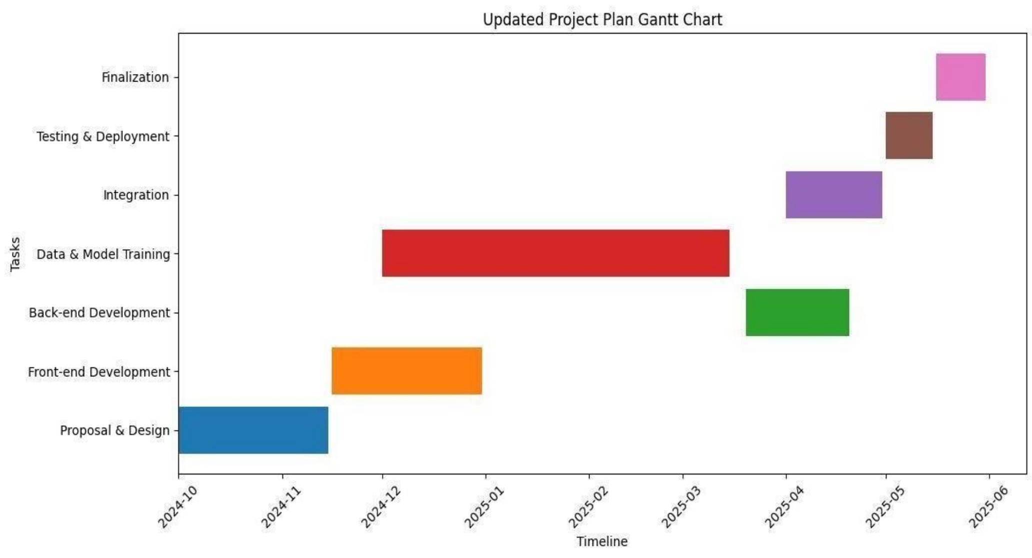


Figure 3. Chant chart

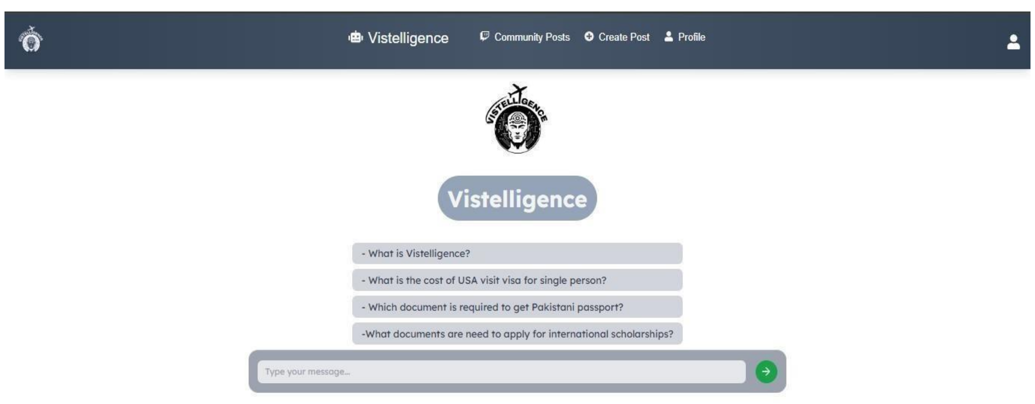
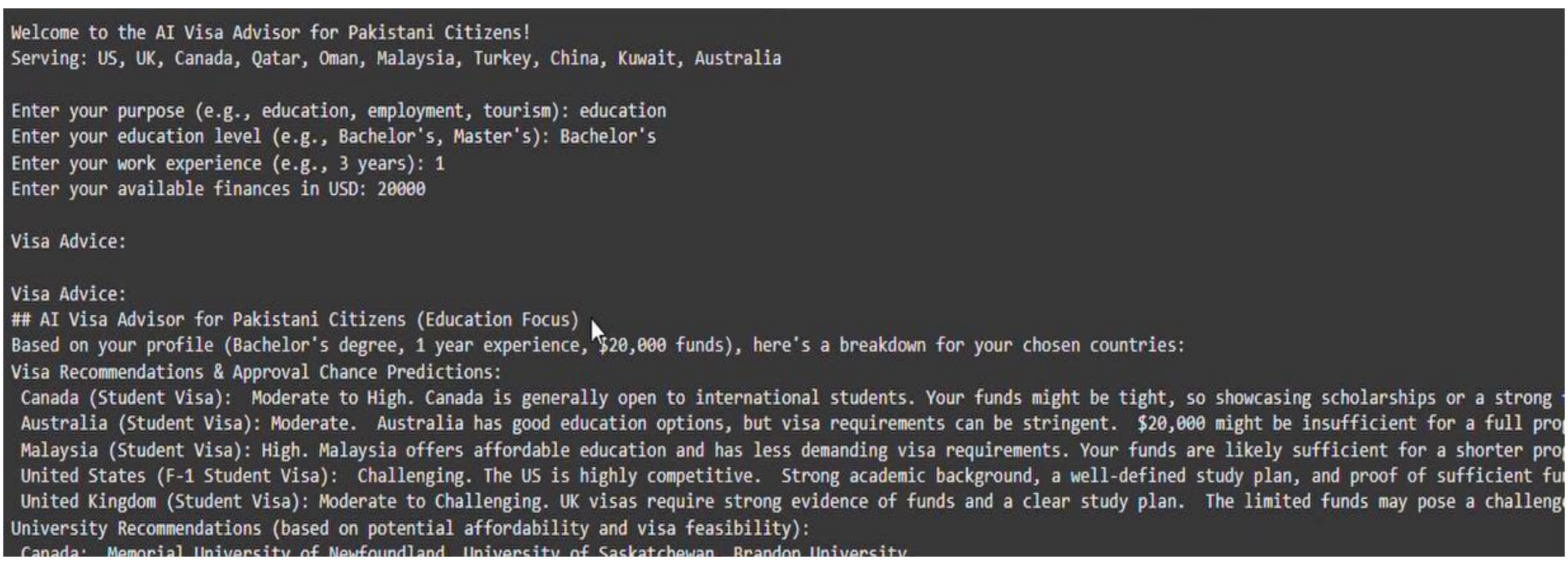


Figure 4. Dashboard (AI model)

Conclusion

Vistelligence makes the visa process easier by using AI to give smart, personalized suggestions. With features like visa prediction, interview tips, and a helpful community, users get the support they need at every step. Our goal is to build a trusted platform that helps people apply for visas with more confidence and less stress.



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Welcome to the AI Visa Advisor for Pakistani Citizens!
Serving: US, UK, Canada, Qatar, Oman, Malaysia, Turkey, China, Kuwait, Australia

Enter your purpose (e.g., education, employment, tourism): education
Enter your education level (e.g., Bachelor's, Master's): Bachelor's
Enter your work experience (e.g., 3 years): 1
Enter your available finances in USD: 20000

Visa Advice:

Visa Advice:
## AI Visa Advisor for Pakistani Citizens (Education Focus)
Based on your profile (Bachelor's degree, 1 year experience, $20,000 funds), here's a breakdown for your chosen countries:
Visa Recommendations & Approval Chance Predictions:
 Canada (Student Visa):  Moderate to High. Canada is generally open to international students. Your funds might be tight, so showcasing scholarships or a strong
 Australia (Student Visa): Moderate.  Australia has good education options, but visa requirements can be stringent.  $20,000 might be insufficient for a full prog
 Malaysia (Student Visa): High. Malaysia offers affordable education and has less demanding visa requirements. Your funds are likely sufficient for a shorter prog
 United States (F-1 Student Visa):  Challenging. The US is highly competitive.  Strong academic background, a well-defined study plan, and proof of sufficient fu
 United Kingdom (Student Visa): Moderate to Challenging. UK visas require strong evidence of funds and a clear study plan.  The limited funds may pose a challenge
University Recommendations (based on potential affordability and visa feasibility):
 Canada:  Memorial University of Newfoundland, University of Saskatchewan, Brandon University
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Figure 5. AI model output sample

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